

Question 1:

Install the Apache web server on your Linux machine using the package manager (apt or yum), and confirm the installation by running the command to check its version.

Answer:

- **For Debian/Ubuntu systems:**
 - Update package index: `sudo apt update`
 - Install Apache: `sudo apt install apache2 -y`
 - Verify version: `apache2 -v`
- **For RHEL/CentOS/Fedora systems:**
 - Install Apache: `sudo yum install httpd -y`
 - Verify version: `httpd -v`

Question 2:

Start the Apache service using `systemctl`, then use `systemctl status` to verify it is running.

Answer:

- **Start the service:**
 - Debian/Ubuntu: `sudo systemctl start apache2`
 - RHEL/CentOS: `sudo systemctl start httpd`
- **Check status:**
 - Debian/Ubuntu: `sudo systemctl status apache2`
 - RHEL/CentOS: `sudo systemctl status httpd`
- **Expected Output:**

The output will display `Active: active (running)` highlighted in green, indicating that the service is running properly.

Question 3:

Stop and restart the Apache service using `systemctl`, and observe the difference between stop, start, and restart commands.

Hint: Use 'systemctl stop apache2', 'systemctl start apache2', and 'systemctl restart apache2' (or 'httpd' on some systems).

Answer:

- **Commands executed:**

- **Stop:** `sudo systemctl stop apache2` (halts the running daemon, making the web server unreachable; status changes to inactive (dead)).
- **Start:** `sudo systemctl start apache2` (launches the service process fresh into memory; status transitions to active (running)).
- **Restart:** `sudo systemctl restart apache2` (shuts down the active processes and immediately launches them again, which is necessary after modifying core configuration files).

Question 4:

Open your browser and access `http://localhost` or your Linux machine's IP address to verify that the Apache default web page is displayed.

Answer:

- **Verification Steps:**

1. Open a web browser on the local machine and navigate to `http://localhost` or `[http://127.0.0.1](http://127.0.0.1)`.
2. If accessing remotely, find the machine's IP address using `ip a` or `hostname -I` and navigate to `http://<YOUR_SERVER_IP>`.
3. The browser will render the default "Apache2 Ubuntu Default Page" (or "HTTPD Test Page"), confirming that the web server is correctly bound to port 80 and serving HTTP traffic.

Question 5:

Write a short comparison (3-4 lines) between Apache and Nginx web servers, mentioning one key use-case for each based on your research.

Answer:

Apache uses a process-driven architecture where each connection creates a new thread, making it highly modular and ideal for dynamic content via `.htaccess` files (e.g., traditional shared hosting platforms). In contrast, Nginx uses an asynchronous, event-driven architecture that handles thousands of concurrent connections with minimal memory usage, making it ideal as a high-performance reverse proxy and load balancer for high-traffic static websites.